

7/22/23, 1:58 AM

CodeProjectFileEditFindViewToolsEducationHelpConfigure...Project Index (read)Configure...

imageGenpy x
12 def generate_base_image(prompt):
13 response = openai.Image.create(
14 prompt=prompt,
15 n=1,
16 size="512x512"
17)
18 return response['data'][0]['url']
19
20 base_image_url = generate_base_image('red apple')
21 img_data = requests.get(base_image_url).content
22 with open('image_name.jpg', 'wb') as handler:
23 handler.write(img_data)
24
25 # Manipulate the image properties
26 def manipulate_image(image, size, aspect_ratio, brightness, contrast, saturation, hue):
27 # Resize the image
28 new_size = (int(image.width * size), int(image.height * aspect_ratio * size))
29 resized_image = image.resize(new_size, Image.ANTIALIAS)
30
31 # Adjust the image properties
32 from PIL import ImageEnhance, ImageOps
33 enhanced_image = ImageEnhance.Brightness(resized_image).enhance(brightness)
34 enhanced_image = ImageEnhance.Contrast(enhanced_image).enhance(contrast)
35 enhanced_image = ImageEnhance.Color(enhanced_image).enhance(saturation)
36
37 # Adjust hue using ImageOps module
38 enhanced_image = ImageOps.colorize(enhanced_image.convert('L'), 'black', 'white', midpoint=128)
39
40 return enhanced_image

188% (40/126)

image_name.jpg | manipulated... x



Code - Manipulating Images

Guide x

Collapse

Image Manipulation

Image Manipulation Function

We first generated a base image of a "red apple" using the DALL-E 2 API. We saved our image as `image_name.jpg`. I know really creative. Now, we are going to access that image and start manipulating it. The first step we are going to do is to create a variable that points to our first image and calls that `base_image`.

```
"""  
base_image_url = generate_base_image('red apple')  
"""  
base_image = Image.open('image_name.jpg')
```

TRY IT

Command was successfully executed.

Click here to refresh your base image

Now we are ready to create a function that will interact with the `PIL` library. The function takes a couple of variables as arguments. Let's start by defining what those are.

```
def manipulate_image(image, size, aspect_ratio, brightness, contrast, saturation, h
```

1. **image**: A PIL image object representing the base image that you want to manipulate.
2. **size**: A float value between 0 and 1, representing the scale factor by which you want to resize the image. A value of 1 will keep the original size, while a value of 0.5 will reduce the size by half, and a value of 0.75 will resize the image to 75% of its original size.
3. **aspect_ratio**: A float value representing the aspect ratio for resizing the image. A value of 1 will maintain the original aspect ratio, while values greater than 1 will stretch the image vertically and values less than 1 will stretch the image horizontally.
4. **brightness**: A float value representing the factor by which to adjust the image's brightness. A value of 1 means no change, while values greater than 1 will increase brightness, and values less than 1 will decrease brightness.
5. **contrast**: A float value representing the factor by which to adjust the image's contrast. A value of 1 means no change, while values greater than 1 will increase contrast, and values less than 1 will decrease contrast.
6. **saturation**: A float value representing the factor by which to adjust the image's color saturation. A value of 1 means no change, while values greater than 1 will increase saturation, and values less than 1 will decrease saturation.
7. **hue**: A float value representing the amount by which to adjust the image's hue. A value of 0 means no change, while positive and negative values will rotate the hue in different directions.

```
# Manipulate the image properties  
def manipulate_image(image, size, aspect_ratio, brightness, contrast, saturation, h  
    # Resize the image  
    new_size = (int(image.width * size), int(image.height * aspect_ratio * size))  
    resized_image = image.resize(new_size, Image.ANTIALIAS)  
  
    # Adjust the image properties  
    from PIL import ImageEnhance, ImageOps  
    enhanced_image = ImageEnhance.Brightness(resized_image).enhance(brightness)  
    enhanced_image = ImageEnhance.Contrast(enhanced_image).enhance(contrast)  
    enhanced_image = ImageEnhance.Color(enhanced_image).enhance(saturation)  
  
    # Adjust hue using ImageOps module  
    enhanced_image = ImageOps.colorize(enhanced_image.convert('L'), 'black', 'white  
    return enhanced_image
```

TRY IT

Command was successfully executed.

We define a function `manipulate_image()` that takes in the base image and various parameters for image manipulation. We use the Python Imaging Library (PIL) to resize the image, and adjust its brightness, contrast, saturation, and hue. We use `ImageOps.colorize()` to adjust the hue of the image.

```
# Example of image manipulation  
manipulated_image = manipulate_image(  
    base_image, size=0.5, aspect_ratio=1, brightness=1.2,  
    contrast=1.5, saturation=0.8, hue=0.1  
)
```

TRY IT

Command was successfully executed.

Finally, we demonstrate how to use this function to create a manipulated version of the original image by adjusting the parameters.

Now we are going to save our manipulated image as `manipulated_red_apple.jpg`.

```
# Save the manipulated image to a file  
manipulated_image.save('manipulated_red_apple.jpg')
```

TRY IT

Command was successfully executed.

Click here to open `manipulated_red_apple.jpg`

What method is used to adjust the hue of the image in the `manipulate_image` function?

☒ ImageOps.colorize
☐ ImageEnhance.Brightness
☐ ImageEnhance.Contrast
☐ ImageEnhance.Color

The `ImageOps.colorize` method is used to adjust the hue of the image in the `manipulate_image` function. The method takes in the converted grayscale image, 'black' and 'white' as color values, and the midpoint adjusted by the hue parameter to rotate the hue in different directions.

Check It

Next